Practical No : 5

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Q.5) write a C program for Stassen's Matrix Multiplication.

```
#include <stdio.h>
int main()
{
   int A[2][2], B[2][2], C[2][2];
   int M1, M2, M3, M4, M5, M6, M7;
   printf("Enter elements of Matrix A:\n");
   for (int i = 0; i < 2; i++)
      for (int j = 0; j < 2; j++)
         scanf("%d", &A[i][j]);
   printf("Enter elements of Matrix B:\n");
   for (int i = 0; i < 2; i++)
      for (int j = 0; j < 2; j++)
         scanf("%d", &B[i][j]);
   // Strassen's seven multiplications
   M1 = (A[0][0] + A[1][1]) * (B[0][0] + B[1][1]);
   M2 = (A[1][0] + A[1][1]) * B[0][0];
   M3 = A[0][0] * (B[0][1] - B[1][1]);
   M4 = A[1][1] * (B[1][0] - B[0][0]);
   M5 = (A[0][0] + A[0][1]) * B[1][1];
   M6 = (A[1][0] - A[0][0]) * (B[0][0] + B[0][1]);
   M7 = (A[0][1] - A[1][1]) * (B[1][0] + B[1][1]);
   // Calculate result matrix
   C[0][0] = M1 + M4 - M5 + M7;
   C[0][1] = M3 + M5;
   C[1][0] = M2 + M4;
   C[1][1] = M1 - M2 + M3 + M6;
```

```
    printf("\nResult Matrix:\n");
    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 2; j++)
            printf("%d ", C[i][j]);
        printf("\n");
    }
    return 0;
}
```

Output

